

Vaishnavi V

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Summary

Computer Science Engineering student with hands-on experience developing full-stack and database-driven applications using React, JavaScript, Python, Flask, MongoDB, and SQL. Strong foundation in software development, problem-solving, and API-based application development. Seeking a Product Engineering Intern role to contribute to scalable software products and gain hands-on industry experience.

Education

B.E. in Computer Science and Engineering

Shri Madhwa Vadiraja Institute of Technology and Management, Bantakal
CGPA: 7.62 | 2023 - Present

12th Standard (PCMB)

Vidyodaya PU College
88.33% | 2023

10th Standard

Vidyodaya Public School
90.33% | 2021

Skills

- **Programming Languages:** Python, Java, C
- **AI:** AI, ML, NLP, GenAI
- **Libraries:** OpenCV, Tesseract OCR, YOLO
- **Web Technologies:** HTML, CSS, JavaScript, React.js, Next.js, Tailwind CSS, TypeScript
- **Database:** SQL (MySQL), MongoDB
- **Backend:** Flask, REST APIs
- **Tools:** Git, GitHub, Docker, VS Code.

Projects

AI-Based Smart Traffic Congestion Optimization

- **Technologies:** Python, OpenCV, YOLOv8, Flask, React.js, TensorFlow, MongoDB
- Developed an AI-powered traffic management system across 4 roads using YOLOv8, OpenCV, Flask, and React.js for real-time vehicle detection and traffic analysis.
- Implemented an LSTM-based traffic prediction pipeline using 90 days of historical data with TensorFlow to forecast traffic patterns.
- Developed adaptive traffic signal optimization to dynamically adjust signal timings based on traffic density and improve traffic flow.

Accident Alert & Emergency Response System

- **Technologies:** Java, Tesseract, GPS/Geo-tagging, MongoDB, Twilio.
- Developed an AI-powered accident management system that uses OCR-based number plate recognition and geo-tagging to detect accidents, identify vehicles, and automatically notify nearby hospitals and authorities for faster emergency response.

Criminal Record Management System

- Developed a role-based full-stack web application for secure criminal record management with authenticated user access.
- Implemented CRUD operations, record search, role-based authorization, and responsive UI for efficient data management.
- **Technologies:** React.js, Next.js, TypeScript, Tailwind CSS, Python, Flask, MongoDB, Axios.

Certificates

- AI Literacy – IBM SkillsBuild
- Hands-on workshop on Git and GitHub
- Webworks workshop on HTML, CSS, JavaScript

Languages

English Kannada Hindi

